Mohammad Karimi

Senior AI Engineer at Proofpoint

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Highlights

- Skilled Senior Data Scientist and ML developer having 8+ years of experience in designing and developing end-to-end ML and data science pipelines
- Expert in high-performance LLM inference, specializing in maximizing GPU compute saturation and memory bandwidth through quantization, continuous batching, and kernel optimizations to deliver peak token throughput.
- Demonstrated strong leadership, representation, and teamwork by collaborating with diverse clients, eagerly tackling new challenges and adapting flexibly to evolving project demands
- Achieved 20+ advanced cloud certifications including GCP Professional ML Engineer and Azure AI Engineer

Relevant Work Experience

Senior AI Engineer Jul. 2025 - Present

Proofpoint, Toronto, ON

- Architected and deployed a high-performance Redaction API on **Kubernetes** using a hybrid of **BERT** transformers and rule-based patterns, delivering a **122x cost reduction** compared to AWS Guardrail while maintaining high precision
- Deployed **20+ open-weight models** on Kubernetes using **vLLM** for classification and reasoning tasks; enabled **KV Caching** and CPU offloading to achieve **10x lower inference costs** than AWS Bedrock
- Optimized **AWS Nova** models on Bedrock for classification by integrating tool use (function calling) and GEPA, increasing the F1-score on the overall dataset by **+40%**
- Developed and fine-tuned **Longformer** architectures for document classification, achieving a **+90% F1-score** and enabling Integrated Gradients for model explainability
- Engineered robust, light-weight classification pipelines using **SentenceBERT**, **Model2Vec**, and MLP classifiers, securing an **82% F1-score** on the hold-out set for resource-constrained environments

Senior Data Scientist Jun. 2023 - Jul. 2025

KPMG Lighthouse, Toronto, ON

- Developed an AI-driven SOC report **generator** using **NLP**, **Transformers**, and local **LLMs** with instruction fine-tuning and inference acceleration, achieving over **92%** accuracy in information retrieval on **GCP**
- Contributed to the development and orchestration of **nine data science pipelines** for **Bell** on **Vertex AI** to predict customer churn, encompassing processes from **data ingestion** to **model deployment** to predict batch loads of recurrent data by using **Cloud Composer**, **PubSub**, **BigQuery**, and many more GCP services
- Designed an end-to-end **ML pipeline** in Python as an **accelerator** for KPMG Lighthouse on GCP

Associate Machine Learning Developer Sep. 2022 - May 2023

AltaML, Edmonton, AB

- Developed and deployed a bias and hate speech classifier based on **Open AI's GPT 3.5 Turbo API** on **Azure** to classify **hate speech** and different types of **biases** from social media posts

- Implemented and deployed a quick and **scalable ML pipeline** on Azure for **Government of Alberta** to predict on over **2.6 billion** geospatial satellite imagery data points with an accuracy of +**88%** using cross-validation

Data Scientist and NLP Researcher Jan. 2021 – Aug. 2022
University of Alberta, Edmonton, AB

- Developed and fine-tuned **language models** (i.e., n-grams and recurrent neural networks) to identify grammatically incorrect writing from **big data** resources with an accuracy of +**97%**
- Implemented a data and ML pipeline to process and predict the **price** of airbnb listings using **XGBoost** with a negligible error rate

Software Engineer Sep. 2018 – Jan. 2020
Iranian national robotics team, Tehran, Iran

- Implemented a data acquisition pipeline to collect imagery dataset of objects and employed **Faster R-CNN** deep learning algorithm to enable object recognition

Education

M.Sc. in Computer Science Jan. 2021 – Aug. 2022
University of Alberta *Edmonton, AB*

B.Sc. in Software Engineering Sep. 2015 – Sep. 2019
University of Science and Technology of Mazandaran *Mazandaran, Iran*

Technical & Soft Skills

Languages: Python, Java, C++, SQL, C#, JavaScript, PHP, HTML/CSS
NLP: NLTK, Spacy, Gensim, Hugging Face, vLLM, Model2Vec, LangChain, DSPy, Unsloth
Libraries: Tensorflow, Pytorch, Keras, Sklearn, OpenCV, YOLO, Pandas, Numpy
Tools: AWS, GCP, Azure, Kubernetes, Docker, Linux, Git, Shell, Bash, Airflow
Database: MySQL, Microsoft SQL Server, SQLite, PostgreSQL
Visualizations: Tableau, Matplotlib, Seaborn, Plotly, Microsoft Power BI, LIME, Streamlit
Soft Skills: Management & Leadership, Critical Thinking, Communication, Work Ethic, Time Management, Creativity

Selected Publications

- **Karimiabdolmaleki, M.**, Farias Wanderley, L., Cutumisu, M. et al. Negative Language Transfer Identification in the English Writing of Chinese and Farsi Native Speakers. Int J Artif Intell Educ (2025). https://doi.org/10.1007/s40593-025-00468-8
- **Karimi Abdolmaleki, M.**, Demmans Epp, C., Cutumisu, M. (2022). Automated feedback generation in programming environments: A systematic review. Poster. Systematic Review and Meta-Analysis SIG Poster Session. American Educational Research Association (AERA) Annual Meeting, San Diego, CA, April 22-25.

Community Engagements

ML Instructor **Alberta AI Association**

- Teaching an advanced ML course with more than 15 students, covering NLP topics and MLOps, including data processing, model training, model evaluation, and deployment

Guest Speaker **Action for Healthy Communities**

- Demonstrated leadership and presentation skills by presenting ML-related topics to people from different backgrounds